

MorphoCLIP: Text-Supervised Contrastive Learning for Perturbation Matching in Cell Painting Images

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Abstract—Cell Painting images record how cells respond to a chemical or genetic perturbation, and matching an image back to the perturbation that produced it is a basic step in phenotypic drug discovery that current methods still do poorly. We present MorphoCLIP, a contrastive model that places Cell Painting wells and text descriptions of perturbations in a shared 512-dimensional space. Each fluorescence channel passes through a frozen DINOv3 ViT-L/16, a one-layer CrossChannelFormer merges the five channel tokens, and a frozen BioClinical ModernBERT encodes the prompt. The channel transformer, two projection heads and one logit-scale scalar are trained, about 14M parameters in total, and one model covers compounds, CRISPR knockouts and ORF overexpressions. Ranking held-out perturbation profiles against 98 candidate texts gives Recall@10 near 39% in both directions on the validation split (random: 10%), and 37% image-to-text and 44% text-to-image on the test split (random: 12%). We test a replicate-alignment image loss, gene-aware soft labels and per-plate offsets measured against replicate plates. Their retrieval differences are unresolved at one seed. The replicate loss raises deterministic replicability mAP on all twelve standard CPJUMP1 tracks, from a pooled 0.298 to 0.343; this gain is partly definitional because the loss directly optimizes replicate proximity. Plate offsets alone do not improve this metric. In separate runs of the same stochastic harness, MorphoCLIP and the released CellCLIP checkpoint give numerically comparable compound replicability fractions, but the runs do not support a ranking. The four-track mean compound target-matching fraction is within the published CellProfiler range, although one track exceeds that range. Gene-compound fraction retrieved is at or near zero for every variant (at most three admitted pairs pass the significance gate), which remains an open failure. We also document an evaluation pitfall: ranking wells against a text pool in which each text appears once per replicate well caps recall and made image-to-text retrieval look worse than chance.

Index Terms—contrastive learning, cell painting, morphological profiling, CLIP, perturbation matching

I. INTRODUCTION

Phenotypic drug discovery asks how a chemical or genetic perturbation changes the state of a cell, at a scale that target-specific assays cannot reach. Bringing one drug to approval costs on the order of \$2 B and takes about a decade [1], while roughly nine of ten compounds that enter clinical trials fail [2].

High-content microscopy is one way to get that scale. The Cell Painting assay [3], [4] stains cells with five fluorescent dyes that mark DNA, endoplasmic reticulum, RNA and nucleoli, actin and Golgi, and mitochondria, and images them as a five-channel picture of cellular state. A perturbation changes

that picture, and the change carries information about the perturbation’s mechanism. The CPJUMP1 benchmark [5] turns this into a retrieval problem over about 3 million fields of view: given images of cells treated with 303 compounds, 160 CRISPR knockouts and 176 ORF overexpressions, can a model match perturbations that act on the same target?

Existing methods each cover part of this problem, and none covers all of it. The CellProfiler pipeline [6] measures handcrafted features and recovers 5 to 25% of the expected sister-perturbation matches on CPJUMP1 [5]. CLOOME [7] and MolPhenix [8] learn contrastive image encoders against molecular fingerprints, and so handle compounds only. CellCLIP [9] replaces fingerprints with text prompts, which admits genetic perturbations, but runs on a 1.48B-parameter DINOv2-g stack and its cross-class retrieval trails CellProfiler. CWA-MSN [10] attacks batch effects directly and beats CellCLIP on gene–gene retrieval with 22M parameters, but has no text branch and cannot be queried in language.

MorphoCLIP combines text supervision, joint training over compounds and genes, and a batch-correction option in one model. It follows CLIP [11]: an image encoder and a text encoder are trained so that a Cell Painting well and the description of its perturbation land close together in a shared 512-dimensional space. Both backbones are frozen and their outputs are cached. The trainable components are a one-layer transformer over channel tokens, two projection heads and one logit-scale scalar, about 14M parameters in total, compared with CellCLIP’s 1.48B total parameters.

The paper makes four contributions.

- 1) A text-supervised model over cached DINOv3 channel tokens [12], trained jointly on compounds, CRISPR knockouts and ORF overexpressions with two frozen backbones.
- 2) Three additions measured separately against a shared control: an image–image replicate loss, gene-aware soft labels, and a per-plate offset relative to replicate plates of the same condition. The replicate loss increases deterministic replicability mAP on every benchmark track; retrieval differences and the other additions remain unresolved at one seed.
- 3) Perturbation-level retrieval with analytic random baselines and evaluation on the standard CPJUMP1 benchmark. Separate stochastic benchmark runs show nu-

merically comparable compound replicability fractions for MorphoCLIP and the released CellCLIP checkpoint, without supporting a ranking. The four-track mean compound target-matching fraction lies within the published CellProfiler range, although one track is above it. Gene-compound fraction retrieved remains at or near zero.

- 4) A documented evaluation pitfall: duplicating each text once per replicate well creates tied blocks of candidates, caps Recall@5 and Recall@10 near Recall@1, and invalidates a shuffled-well baseline for image-to-text retrieval.

II. PROBLEM STATEMENT

Given a Cell Painting image $\mathbf{x} \in \mathbb{R}^{5 \times H \times W}$ (five fluorescence channels, acquired at 1080×1080 pixels and resized to the backbone processor’s native 224×224 input) and a text description t of the perturbation applied to the imaged cells, we learn encoders f_{img} and f_{txt} that map both into a shared unit sphere:

$$f_{\text{img}}(\mathbf{x}), f_{\text{txt}}(t) \in \mathbb{R}^d, \quad \|f_{\text{img}}(\mathbf{x})\|_2 = \|f_{\text{txt}}(t)\|_2 = 1. \quad (1)$$

Matching image–text pairs should have high cosine similarity and non-matching pairs low similarity. We use $d = 512$. In practice the image unit is a well: all sites of one well are encoded together and produce one vector.

The trained encoders support three retrieval tasks:

- **Text-to-image retrieval.** Given a perturbation description, retrieve the wells that show the corresponding phenotype.
- **Image-to-text retrieval.** Given a well, return the perturbation description that best matches its morphology.
- **Image-to-image retrieval.** Compare well embeddings by cosine similarity, without text at inference time. This is the setting of the standard CPJUMP1 benchmark: do replicate wells of one perturbation match, and do perturbations that share a target match across modalities?

Three properties of the data make this harder than natural image–caption alignment.

a) Batch effects.: The same perturbation looks different across plates and imaging days. Illumination, staining and focus drift add technical variance that is often larger than the biological signal. In CPJUMP1, replicability depends on modality (compounds replicate more cleanly than CRISPR) and on cell line (U2OS against A549) [5], and the distribution of per-perturbation average precision is heavily right-skewed: most perturbations score close to zero.

b) Fuzzy ground truth.: Perturbations do not come with clean positive and negative labels. Two compounds may share a primary target and diverge on secondary effects. A CRISPR knockout and a small molecule can converge on one pathway through different mechanisms. Similarity is a continuum, which argues for soft-positive losses over hard-negative InfoNCE.

c) Scale and compute.: CPJUMP1 has 51 plates and 3.31 TiB of raw 16-bit TIFF images, about 3 million fields of view and 75 million cells. Training full vision and language encoders end to end at this scale is out of reach on one consumer GPU. We instead run the frozen backbones once, cache their outputs, and train only the small components that sit on top.

III. RELATED WORK

A. Morphological Profiling

Cell Painting [3] fixed the five-stain protocol that most high-content morphological profiling now uses [4]. The classical analysis tool, CellProfiler [6], measures more than 1,000 handcrafted features per cell. On CPJUMP1 [5] that pipeline retrieves 5 to 25% of the expected sister perturbations within a modality and 7 to 17% of CRISPR guide matches, and cross-modal compound-to-gene matching is harder still.

Handcrafted pipelines encode a fixed set of measurements. Self-supervised vision transformers trained with DINO outperform CellProfiler on drug-target and gene-family classification with zero-shot transfer [13], motivating our use of frozen foundation-model features.

B. Contrastive Learning for Cell Painting

CLOOME [7] was the first CLIP-style model for Cell Painting. It trains a ResNet image encoder against a Morgan-fingerprint MLP and reaches top-1 retrieval more than $70 \times$ random over about 2,000 candidates. It handles chemical perturbations only and treats the five channels as RGB.

MolPhenix [8] swaps the trained image branch for a frozen Phenom-1 encoder, averages replicate embeddings, and adds a soft-to-label loss, gaining $8.78 \times$ over CLOOME. It also covers compounds only and depends on a proprietary backbone.

CellCLIP [9] is the closest prior work. It uses templated prompts (“A Cell Painting image of [cell_type] cells treated with [drug_name], SMILES: [SMILES_string]”), a Cross-ChannelFormer that encodes each channel separately before attending across channels, attention-based pooling across cells, and a Continuously Weighted Contrastive Loss (CWCL) for soft positives. It runs on DINOv2-g (1.1B parameters, 1.48B in total) and is evaluated on the Bray dataset [14] and on RxRx3-core gene–gene benchmarks (CORUM recall 0.354). Its text branch admits every perturbation type, but its cross-class mAP falls below the CellProfiler baseline, so text supervision alone does not solve the batch-effect problem.

C. Batch-Effect Correction

CWA-MSN [10] addresses batch effects inside a masked siamese network by aligning embeddings of cells that received the same perturbation across wells and plates. It absorbs technical variance without an explicit batch label. With 22M parameters and 0.2M training images it improves on CellCLIP by 9% and on OpenPhenom by 29% on gene–gene retrieval (CORUM recall 0.386). It has no text branch.

TABLE I

CELL PAINTING REPRESENTATION LEARNING METHODS. THE PARAMETER COUNT IS TRAINABLE PARAMETERS FOR MORPHOCLIP AND TOTAL PARAMETERS FOR THE OTHER LEARNED METHODS.

Method	Image Encoder	Text	Pert. Types	Batch Corr.	Params
CellProfiler [6]	Handcrafted	No	All	Sphering	N/A
CLOOME [7]	ResNet	No	Compounds	No	~25M
MolPhenix [8]	Phenom-1	No	Compounds	No	~36M
CellCLIP [9]	DINOv2-g	Yes	All	No	1,477M
CWA-MSN [10]	ViT-S/16	No	Unimodal	CWA	22M
MorphoCLIP	DINOv3-L	Yes	All	Plate offsets	~14M

D. Foundation Models for Microscopy

DINOv2 [15] and DINOv3 [12] give self-supervised visual features that transfer to biological imaging despite natural-image pretraining [13]. DINOv3 adds gram anchoring and a revised training recipe. Channel Vision Transformers [16] show that per-channel encoding followed by cross-channel attention works for multi-channel microscopy, which is the pattern our channel aggregator follows.

On the text side, BioClinical ModernBERT [17] (150M parameters) is pre-trained on 53.5B tokens of PubMed, PMC and clinical text, about $17\times$ the biomedical corpus behind the PubMedBERT (110M, 3.1B tokens) that CellCLIP uses. It is built on ModernBERT (rotary embeddings, Flash Attention, GeGLU) and can encode both small-molecule and gene descriptions with one model, unlike SMILES-specialized encoders such as ChemBERTa.

E. Positioning of MorphoCLIP

Table I places MorphoCLIP among these methods. CellCLIP has text but no batch correction. CWA-MSN has batch correction but no text. MorphoCLIP has both, trains on all three perturbation types, and keeps the trained part small. Where a like-for-like number exists we compare on the same benchmark harness (Section VI); elsewhere we quote published figures and say what differs.

IV. METHOD

MorphoCLIP has image and text encoders trained with a contrastive text objective and optional replicate and plate-offset terms. The image encoder turns a well of five-channel Cell Painting sites into a 512-dimensional unit vector. The text encoder does the same for a prompt that describes the perturbation. Figure 1 shows the pipeline.

A. Image Encoder

A Cell Painting site has five fluorescence channels (Table II). We encode each channel on its own: the single-channel image is resized to the Hugging Face processor’s native 224×224 input, copied to three channels, normalized with the processor statistics, and passed through a frozen DINOv3 ViT-L/16 [12] (300M parameters). The result is five 1024-dimensional CLS tokens per site,

$$\{\mathbf{z}_c\}_{c=1}^5, \quad \mathbf{z}_c \in \mathbb{R}^{1024}. \quad (2)$$

TABLE II

CELL PAINTING FLUORESCENCE CHANNELS USED IN MORPHOCLIP.

Ch	Stain	Target
1	MitoTracker / Alexa 647	Mitochondria
2	Phalloidin / Alexa 568	Actin
3	WGA / Alexa 488	Golgi / plasma membrane
4	Concanavalin A / Alexa 488	Endoplasmic reticulum
5	Hoechst 33342	DNA / nucleus

We use DINOv3 ViT-L rather than the larger DINOv2-g backbone used by CellCLIP. The channel tokens are written to disk once, and every training run reads from that cache. The backbone never runs during training.

a) *CrossChannelFormer (CCF)*.: A one-layer transformer encoder [18] with 4 heads merges the five channel tokens into one site vector, following the channel-aware design of CellCLIP [9] and Channel ViT [16]. Each token is L2-normalized so the five channels enter at a common scale, then a learned channel embedding $\mathbf{e}_c$ is added so the transformer can tell the stains apart. A learned aggregation token $\mathbf{q}$ is prepended and its output is the site vector:

$$\mathbf{h}_{\text{site}} = [\text{CCF}(\mathbf{q}, \hat{\mathbf{z}}_1 + \mathbf{e}_1, \dots, \hat{\mathbf{z}}_5 + \mathbf{e}_5)]_0, \quad \hat{\mathbf{z}}_c = \mathbf{z}_c / \|\mathbf{z}_c\|_2. \quad (3)$$

The layer is pre-norm with a feed-forward width of 4096 and GELU activations. The transformer can read out joint patterns across stains, for example a change that shows in both the mitochondrial and the ER channel. The CCF holds 12.6M of the 14M trained parameters.

b) *Site-to-well pooling*.: The training unit is a well. Site vectors are averaged under a site mask, so wells with 7, 9 or 16 imaged sites all produce one vector and the order of sites does not matter.

c) *Image projection head*.: A two-layer MLP with LayerNorm, GELU and dropout ($p = 0.3$) maps the well vector $\mathbf{h}_{\text{img}}$ to the shared space:

$$f_{\text{img}}(\mathbf{x}) = \text{L2Norm}(\mathbf{W}_2 \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_1 \mathbf{h}_{\text{img}})))), \quad (4)$$

with $\mathbf{W}_1 \in \mathbb{R}^{512 \times 1024}$ and $\mathbf{W}_2 \in \mathbb{R}^{512 \times 512}$.

B. Text Encoder

Each perturbation gets one prompt, built from CPJUMP1 metadata and external annotations with one of four templates:

- **Compound**: “Cell Painting morphological profile of $\{\text{cell_line}\}$ cells treated with the compound $\{\text{compound_name}\}$. Chemical structure (SMILES): $\{\text{smiles}\}$. Known target gene: $\{\text{target_gene}\}$. Target protein function: $\{\text{gene_function}\}$. Perturbation modality: chemical compound.”
- **CRISPR**: “... with CRISPR-Cas9 knockout of gene $\{\text{gene_symbol}\}$. Gene description: $\{\text{gene_description}\}$. Gene function: $\{\text{gene_function}\}$. Perturbation modality: CRISPR knockout.”
- **ORF**: “... overexpressing gene $\{\text{gene_symbol}\}$ via open reading frame construct. Gene description:

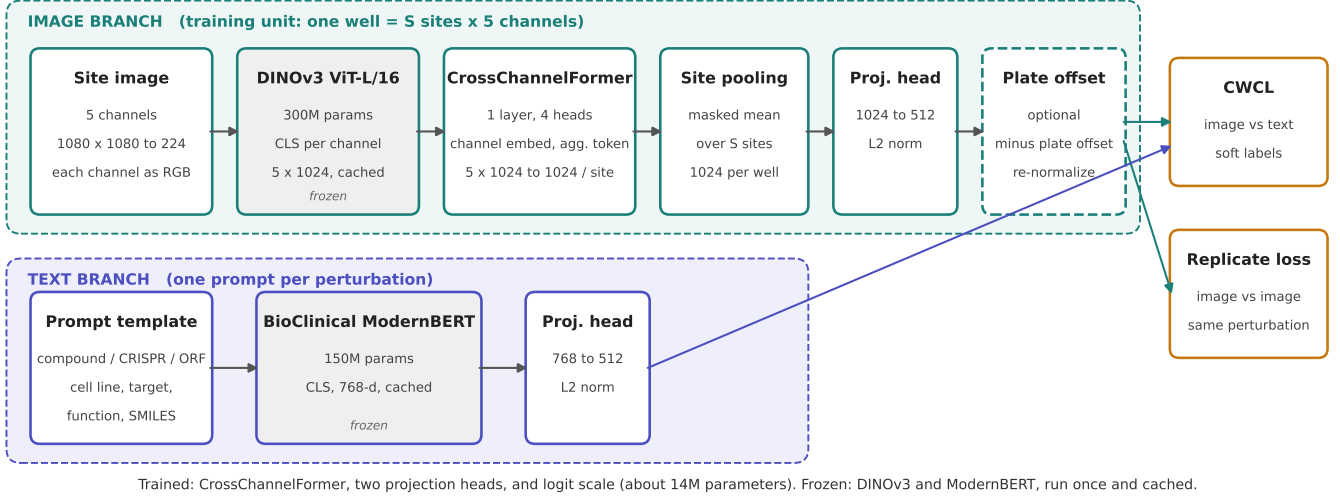


Fig. 1. MorphoCLIP. Each fluorescence channel of a site is encoded on its own by a frozen DINOv3 ViT-L/16, giving five 1024-d CLS tokens. A one-layer CrossChannelFormer attends over the five channel tokens and a learned aggregation token, and the aggregation token’s output is the site vector. Site vectors are mean-pooled into a well vector and projected to the shared 512-d space. The prompt is encoded by a frozen BioClinical ModernBERT and projected to the same space. When plate offsets are on, the image embedding is shifted by its plate’s offset and re-normalized before the losses. The trainable components are the CrossChannelFormer, the two projection heads and one logit-scale scalar, about 14M parameters in total.

$\{gene_description\}$. *Gene function:* $\{gene_function\}$.
Perturbation modality: *ORF overexpression.*

- **Negative control:** “... treated with DMSO vehicle control. No active perturbation applied.”

Missing fields become “unknown”. Compound annotations come from ChEMBL and the CPJUMP1 metadata. Gene descriptions and functions come from UniProt. Prompts are keyed by the perturbation identifier (`broad_sample`), so replicate wells of one perturbation share one text vector.

We encode each prompt with a frozen BioClinical ModernBERT [17] (150M parameters, 768-d CLS token). A trainable projection head with the same shape as the image head maps the CLS token to the shared space:

$$f_{\text{txt}}(t) = \text{L2Norm}(\mathbf{W}_4 \text{Drop}(\text{GELU}(\text{LN}(\mathbf{W}_3 \mathbf{h}_{\text{txt}})))), \quad (5)$$

with $\mathbf{W}_3 \in \mathbb{R}^{512 \times 768}$ and $\mathbf{W}_4 \in \mathbb{R}^{512 \times 512}$. The 768-d BERT outputs are cached separately from the projected 512-d vectors, so a change to the projection head never re-runs BERT.

C. Training Objective

a) *Continuously weighted contrastive loss (CWCL).*: InfoNCE [19] treats every off-diagonal pair in a batch as a negative. That is wrong when two wells in the batch carry the same perturbation, and questionable when two compounds share a target. Following CellCLIP [9], we use a soft-label variant. For a batch of N wells with logits $\mathbf{S} = \tau \mathbf{F}_{\text{img}} \mathbf{F}_{\text{txt}}^\top$, an affinity matrix $\mathbf{W} \in [0, 1]^{N \times N}$ sets $w_{ij} = 1$ when wells i and j carry the same perturbation, $w_{ij} = \alpha$ when their perturbations differ but their target-gene sets overlap, and $w_{ij} = 0$ otherwise. The loss is

$$\mathcal{L}_{\text{CWCL}} = -\frac{1}{2N} \sum_{i=1}^N \sum_{j=1}^N \left(\tilde{w}_{ij} \log p_{ij} + \tilde{w}_{ij}^\top \log p_{ij}^\top \right), \quad (6)$$

where p_{ij} is the softmax of row i of $\mathbf{S}$, $\tilde{w}_{ij} = w_{ij} / \sum_k w_{ik}$, and the transposed terms use $\mathbf{S}^\top$ and $\mathbf{W}^\top$ normalized over their own rows. The gene weight α is a hyperparameter; the base model uses $\alpha = 0$ (binary labels) and the soft-label ablation uses $\alpha = 0.6$. The scale τ is learned through its logarithm, which starts at $\ln(1/0.07)$ and is clamped to $[0, \ln 100]$ after every step.

b) *Replicate alignment loss.*: The text loss never asks two replicate wells of one perturbation to be near each other. It only asks each to be near their shared text. We add an image-image term that does. With P_i the set of other wells in the batch that carry the same perturbation as well i and $\mathbf{G} = \tau \mathbf{F}_{\text{img}} \mathbf{F}_{\text{img}}^\top$ with the diagonal masked out,

$$\mathcal{L}_{\text{rep}} = -\frac{1}{|\{i : P_i \neq \emptyset\}|} \sum_{i: P_i \neq \emptyset} \frac{1}{|P_i|} \sum_{j \in P_i} \log \frac{\exp G_{ij}}{\sum_{k \neq i} \exp G_{ik}}. \quad (7)$$

Wells with no in-batch replicate are dropped from the mean, and the term shares the learned temperature with the text loss. The training loss is $\mathcal{L} = \mathcal{L}_{\text{CWCL}} + \lambda_{\text{rep}} \mathcal{L}_{\text{rep}}$; the base model uses $\lambda_{\text{rep}} = 0$ and the ablation uses 0.3. Validation loss is the text term alone, so model selection is not influenced by the added term.

c) *Perturbation-aware batches.*: For the replicate term to have positives, replicate wells must share a batch. Our sampler groups the training wells by perturbation, cuts each group into chunks of two wells, shuffles the chunks and packs them into batches of 256. A repair pass swaps chunks so that every batch spans at least two plates where possible. This is a batch-composition constraint; it does not guarantee that individual negative pairs are free of plate confounding. Every ablation run uses this sampler; the base model uses uniform random batches.

d) *Plate offsets (cross-well alignment).*: Batch effects are the main confound in Cell Painting, but in CPJUMP1 a plate is close to synonymous with an experimental condition (one cell line, one perturbation modality, one timepoint). Subtracting a plate’s mean embedding would therefore also subtract the condition, which is exactly what the prompt describes. Our correction is relative to replicate plates instead. At the start of each epoch a gradient-free pass over the training wells computes each plate’s mean embedding $\bar{\mathbf{f}}_p$, and every well is shifted by its plate’s offset from the mean of the plates in the same condition:

$$\tilde{\mathbf{f}}_w = \frac{\mathbf{f}_w - \delta_{p(w)}}{\|\mathbf{f}_w - \delta_{p(w)}\|_2}, \quad \delta_p = \bar{\mathbf{f}}_p - \frac{1}{|C(p)|} \sum_{q \in C(p)} \bar{\mathbf{f}}_q, \quad (8)$$

where $p(w)$ is the plate of well w and $C(p)$ is the set of plates that share plate p ’s condition. Offsets within a condition sum to zero, so the step removes replicate-to-replicate drift and cannot remove the condition itself. The condition of each plate comes from the CPJUMP1 reference metadata; the eleven plates that the reference table does not cover (non-standard seeding density, antibiotics, or compounds on the Cas9 line) are keyed on every condition axis from the experiment table so they are not merged into a standard group. Offsets are constants (no gradient flows through them), apply to image embeddings only, are frozen within an epoch, and are stored in the checkpoint so evaluation and profile export apply the same correction as training. An earlier version of this component subtracted the in-batch plate mean instead. It removed the condition signal and cut retrieval to near chance; Section VI reports that run for completeness.

D. Optimization

We train with AdamW [20] at learning rate 10^{-4} , $(\beta_1, \beta_2) = (0.9, 0.999)$ and $\varepsilon = 10^{-8}$. Weight decay 0.2 is applied to parameters with at least two dimensions except parameters whose names identify biases or normalization layers; it is not applied to one-dimensional parameters or the learned temperature. The learning rate warms up linearly for 100 steps and then follows a cosine schedule to zero. We use FP16 mixed precision, gradient clipping at norm 1.0 and a batch of 256 wells. The base model trained on a 100-epoch schedule and its best validation loss came at epoch 9; the ablation runs use a 30-epoch schedule with early stopping after 8 epochs without improvement. The recorded `epoch_seconds` is about four seconds for 55 optimizer steps on an RTX 5080 with the cache in RAM. That timer excludes validation and, for offset runs, the gradient-free offset refresh, so it is not end-to-end epoch or run time.

V. EXPERIMENTAL SETUP

A. Dataset

We use the CPJUMP1 pilot [5], a public dataset from the JUMP Cell Painting Consortium hosted on the Cell Painting Gallery (S3 prefix `cpg0000-jump-pilot`). It has 51 plates and 3.31 TiB of raw 16-bit TIFF images. The perturbations are 303 small-molecule compounds, 160 CRISPR knockouts

and 176 ORF overexpressions, in two cell lines (U2OS and A549), mostly at two timepoints per modality. Each gene target is linked to at least two compounds, which is what makes compound–gene matching testable. Each plate has 384 wells with 7 to 16 imaged sites per well, and each site has five fluorescence channels at 1080×1080 pixels plus three brightfield channels that we do not use.

B. Data Pipeline

The raw plates are 58 to 118 GiB each, so we stream, encode and delete. Each plate is downloaded to scratch, passed once through the frozen DINOv3 backbone, and removed once the per-channel CLS tokens are on disk. The resulting channel-token cache is read by every training run. Text embeddings are cached the same way: each prompt is encoded once by the frozen ModernBERT and the 768-d CLS token stored.

C. Splits

Wells are split by perturbation. Each `broad_sample` is hashed with a fixed seed into train, validation or test at 80/10/10, so all replicate wells of a perturbation land in one split and every split contains all three modalities. Control wells are excluded from training and evaluation. The validation split has 2,220 wells from 98 perturbations (42 compound, 34 CRISPR, 22 ORF), about 24 replicate wells each. The test split has 1,860 wells from 86 perturbations. The hash is deliberately stable: changing it would move every well to a different split and invalidate every split saved on disk.

D. Retrieval Protocol

Replicate wells share one text vector, so the text side of a split has P distinct vectors (98 on validation) and the image side has N wells (2,220). We report four directions, each with an analytic random baseline computed from the same P and N .

- **Well-level image-to-text.** Each of the N wells ranks the P unique texts. One candidate is correct. Random Recall@ k is k/P .
- **Well-level text-to-image.** Each of the P texts ranks all N wells and is scored on the rank of its first correct replicate. With m replicate wells among N candidates, random Recall@ k is $1 - \binom{N-m}{k} / \binom{N}{k}$.
- **Perturbation-level, both directions.** The well embeddings of each perturbation are averaged and re-normalized into one profile, and the P profiles are ranked against the P texts. This is the unit of the standard CPJUMP1 protocol. Random Recall@ k is k/P .

We report Recall@{1, 5, 10} and the median rank of the first correct match. Model selection uses the validation text loss, and we report both the validation and the test split.

An earlier version of this evaluation ranked each well against all N well-texts instead of the P unique texts. Because the pool then held about 24 exact copies of every text vector, a model that scores identical vectors identically placed all copies at consecutive ranks, which capped Recall@5 and Recall@10 near Recall@1 and made image-to-text retrieval

look worse than a shuffled baseline that interleaved the copies. Section VI-E gives the arithmetic. Every current retrieval result uses the unique-text pool.

E. Standard CPJUMP1 Benchmark

We also run the benchmark of Chandrasekaran et al. [5] on exported well embeddings, with the pinned copairs implementation the reference code uses. Trained checkpoints encode every well of a plate (controls included) in fp32, subtract the plate’s negative-control mean, and write one profile CSV per plate in the layout the benchmark reads. The benchmark covers the 40 plates at standard seeding density without antibiotics, at two timepoints per modality (short: compound 24 h, ORF 48 h, CRISPR 96 h; long: 48 h, 96 h, 144 h). It measures three things with average precision (AP) against a permutation null: *replicability*, whether replicate wells of one perturbation retrieve each other; *target matching*, whether perturbations of one modality that share a target retrieve each other; and *gene-compound matching*, whether compounds retrieve the CRISPR or ORF perturbations of their target. Each track reports mean AP (mAP) and the fraction of perturbations whose AP passes $q < 0.05$ (fraction retrieved). Summary fractions are unweighted means over tracks; summary mAP values pool the scored rows and are therefore weighted by track size.

One caveat applies to every fraction-retrieved value in this paper and in the reference results: the permutation null is random, so the set of perturbations that pass the threshold changes from run to run of identical code. Only replicability mAP is deterministic. Small differences in fraction retrieved should not be read as effects.

F. Baselines

- **CellProfiler** [6]. Handcrafted features. On CPJUMP1 target matching, the published fraction retrieved is 4.3 to 25.1% [5]. We did not rerun the CellProfiler pipeline.
- **CellCLIP** [9]. We quote the published Bray-dataset retrieval (best Recall@10 7.4%) and CORUM recall (0.354). We also ran the released CellCLIP checkpoint through our benchmark harness on the short timeline, exporting profiles with CellCLIP’s own encoder path. Those numbers are the closest like-for-like comparison we have, with two caveats noted where they appear: CellCLIP’s evaluation recipe includes a control-fitted KernelPCA batch correction that our export does not apply, and the two are separate runs of a non-deterministic harness.
- **CWA-MSN** [10]. CORUM gene-gene recall 0.386 with 22M parameters. Its benchmarks (CORUM, HuMAP, Reactome) are not among ours, so no comparable number exists.

G. Ablation Design

The base model was trained with binary CWCL labels, uniform random batches, no replicate term and no plate offsets, on the 100-epoch schedule. The ablations share one config

that differs from the base in three ways: perturbation-aware batches, a 30-epoch schedule and early stopping (patience 8). From that shared control we change one thing at a time: gene-aware soft labels ($\alpha = 0.6$), the replicate alignment loss ($\lambda_{\text{rep}} = 0.3$), and plate offsets. The plate-offset variant was run three times: once with the earlier in-batch mean subtraction, once with condition-relative offsets for the 40 plates the reference metadata covers and a zero offset for the other 11, and once with offsets for all 51 plates. A final run turns all three additions on together. Every run uses seed 42, batch size 256 and the same split. All retrieval numbers below therefore come from one seed. At 98 validation candidates one retrieved perturbation changes recall by 1.02 percentage points; at 86 test candidates it changes recall by 1.16 points. Differences of a few points are therefore only a few perturbations and are not treated as effects without repeated seeds.

H. Implementation

Everything runs on one RTX 5080 (16 GB). AdamW, learning rate 10^{-4} , weight decay 0.2, 100 warm-up steps then cosine decay, FP16, gradient clipping at 1.0. CCF: one layer, four heads. Projection heads: hidden width 512, dropout 0.3. The code, configs and split manifests are released with the paper.

VI. RESULTS AND DISCUSSION

All numbers come from a single seed. Retrieval numbers are on the validation and test splits described in Section V-D; benchmark numbers are on the 40 benchmark-eligible plates. We give the random baseline next to every retrieval figure and note where a difference is smaller than one perturbation’s worth of recall.

A. Cross-Modal Retrieval

Table III reports the base model on both splits. On validation, ranking the 98 perturbation profiles against the 98 texts gives Recall@10 of 38.8% image-to-text and 39.8% text-to-image, against a random 10.2%, with a median rank of 13. On the 86 test perturbations the same checkpoint gives 37.2% and 44.2% against a random 11.6%. These recalls are 3.2 to 3.9 times their analytic chance rates.

Well-level text-to-image is the one direction that looks weak in Recall@10 (23.5% against 9.7%), but that metric asks the text to place one of its 24 replicate wells in the top ten of 2,220, and its median rank of 30 says the first replicate is usually found within the top 1.4% of the pool. Once wells are averaged into profiles the two directions agree.

B. Ablations

Table IV lists the single-factor runs.

The retrieval differences among the three additions are unresolved at one seed. The ordering changes on the test split, and changes of several percentage points represent only a few perturbations. We therefore do not claim a retrieval gain for any single addition. The replicate-loss checkpoint is selected at epoch 7 and the control at epoch 11, but one run per condition does not establish a difference in convergence rate.

TABLE III

RETRIEVAL OF THE BASE MODEL ON THE VALIDATION SPLIT (2,220 WELLS, 98 PERTURBATIONS) AND THE TEST SPLIT (1,860 WELLS, 86 PERTURBATIONS). WELL-LEVEL IMAGE-TO-TEXT RANKS THE UNIQUE TEXTS; WELL-LEVEL TEXT-TO-IMAGE RANKS ALL WELLS AND SCORES THE FIRST CORRECT REPLICATE; PERTURBATION-LEVEL RANKS MEAN PROFILES AGAINST TEXTS. RANDOM ROWS ARE ANALYTIC. MEDIAN RANK IS OVER THE CANDIDATE POOL OF EACH DIRECTION (P TEXTS, N WELLS, OR P PROFILES).

Split	Direction	R@1	R@5	R@10	Med. rank
Val	well image $\rightarrow$ text	5.9	23.3	39.4	14
	well text $\rightarrow$ image	4.1	15.3	23.5	30
	pert. image $\rightarrow$ text	4.1	20.4	38.8	13
	pert. text $\rightarrow$ image	5.1	18.4	39.8	13
	random, single positive	1.0	5.1	10.2	49.5
	random, well text $\rightarrow$ image	1.0	5.0	9.7	99
Test	well image $\rightarrow$ text	5.5	22.0	41.4	13
	well text $\rightarrow$ image	4.7	18.6	26.7	25
	pert. image $\rightarrow$ text	4.7	17.4	37.2	14
	pert. text $\rightarrow$ image	8.1	23.3	44.2	13
	random, single positive	1.2	5.8	11.6	43.5
	random, well text $\rightarrow$ image	1.2	5.7	11.0	86

The in-batch offset version yields perturbation-level image-to-text Recall@10 of 15.3%, compared with 37.8% for its control, and its best checkpoint occurs at epoch 4. Evaluating that checkpoint with correction disabled at inference gives 12%, so the degradation arose during training. In CPJUMP1 a plate is nearly one condition, and the plate mean carries modality and cell-line information named by the prompt. The condition-relative offset avoids this degradation. Extending the condition map from 40 to all 51 plates changes validation image-to-text recall by five perturbations, which remains unresolved at one seed.

The combined run is among the higher-scoring rows on both splits but is not uniformly best. Its best validation text loss is 5.396, compared with 5.438 for the control. These single-seed results show that the three additions can be trained together; they do not establish an additive retrieval benefit.

C. Standard CPJUMP1 Benchmark

Table V reports fraction retrieved and mAP on the standard benchmark for the base model, the replicate-loss run, the plate-offset run and the combined run.

a) Replicability.: The replicate loss increases mAP on all twelve tracks; pooled mAP rises from 0.298 to 0.343. This deterministic metric is the clearest change in the study. It is also partly definitional: the loss pulls replicate wells of one perturbation together, and replicability measures whether those wells retrieve one another. The result shows that the loss achieves its direct objective, not that the embedding is more biologically informative. Plate offsets alone give pooled mAP 0.292. The combined run has pooled mAP 0.369 and exceeds the replicate-loss run on ten tracks while being slightly lower on two ORF tracks. Fraction-retrieved values are descriptive because their significance gate is stochastic.

b) Target matching.: The scored population changes with the replicability gate: the base, replicate-loss and combined

runs admit 536, 748 and 856 consensus profiles. The CRISPR tracks contain only 2 to 30 targets, so their stochastic fractions move in large steps. Across the larger compound tracks, no variant is consistently ahead in mAP. The target-matching rows therefore do not establish a downstream benefit from the replicate loss or plate offsets.

c) Gene-compound matching.: Fraction retrieved is at or near zero for every variant. Across the runs, zero to three admitted pairs pass the stochastic significance gate. The nonzero compound-CRISPR pairs have 16 to 33 targets and reach 0.06 at most; no pair is nonzero in more than two runs. Pooled mAP is 0.08 to 0.14, with no consistent ordering. The training changes therefore provide no positive cross-modality result at this seed count.

D. Comparison with Baselines

a) CellProfiler.: The published CellProfiler fraction retrieved on CPJUMP1 target matching is 4.3 to 25.1% [5]. The base model’s four compound target-matching tracks have an unweighted mean of 16.3%, which lies within that range; three individual tracks lie within it and one is higher at 31.5%. We did not rerun CellProfiler, and the harness comparison is not paired, so these values do not establish an advantage. Replicability is a different task and is not comparable with the published range.

b) CellCLIP.: Table VI reports short-timeline fraction retrieved from separate runs of the same harness. CellCLIP’s recipe fits a KernelPCA on control wells before scoring, whereas ours does not, and the significance gate is stochastic. The numerical values differ by track, but these unpaired fractions do not support claims of parity, superiority or inferiority. The table is included as a same-harness reference, not a ranking.

c) CWA-MSN.: CWA-MSN reports CORUM recall of 0.386 on RxRx3-core. We have no number on that benchmark and make no comparison. Its core mechanism, pulling embeddings of one perturbation together across wells, is closer to our replicate alignment loss than to our plate offsets, and it is that mechanism, not the offsets, that moved replicability here.

E. Evaluation Pitfall: Duplicated Candidates

The first version of our evaluation ranked each well against all 2,220 well-texts. Replicate wells share one text vector, so that pool contained 98 distinct vectors, each present about 24 times. A model scores identical vectors identically, so the copies of the correct text occupy consecutive ranks. If the correct text ranks first among the distinct texts, its copies fill ranks 1 to 24, and Recall@5 and Recall@10 collapse to Recall@1. More generally, a unique-text rank of 13 maps to the first duplicate at rank $12 \times 24 + 1 = 289$. The corrected protocol ranks the 98 unique texts. Text-to-image does not have this problem because its candidates are distinct wells. The random baseline must follow the multiplicity of the candidate pool.

TABLE IV

SINGLE-FACTOR ABLATIONS FROM A SHARED CONTROL (PERTURBATION-AWARE BATCHES, 30-EPOCH SCHEDULE, EARLY STOPPING, SEED 42). PERTURBATION-LEVEL RECALL@10 IN PERCENT, MEDIAN RANK OVER P TEXTS, ON VALIDATION (98 PERTURBATIONS) AND TEST (86). RANDOM RECALL@10 IS 10.2 ON VALIDATION AND 11.6 ON TEST. ONE PERTURBATION IS ABOUT ONE POINT. BEST EPOCH IS THE EPOCH SELECTED BY VALIDATION TEXT LOSS.

Run	Change from control	Validation			Test			Best ep.
		i→t	t→i	Med.	i→t	t→i	Med.	
Base model	100-epoch schedule, random batches	38.8	39.8	13	37.2	44.2	14	9
Control	none	37.8	37.8	13	50.0	41.9	10	11
Soft labels	$\alpha = 0.6$	37.8	42.9	14	44.2	47.7	12	13
Replicate loss	$\lambda_{\text{rep}} = 0.3$	44.9	41.8	13	47.7	46.5	12	7
Plate offsets (in-batch, removed)	use_cwa	15.3	10.2	45	—	—	—	4
Plate offsets (40 of 51 plates)	use_cwa	36.7	39.8	13	—	—	—	8
Plate offsets (all 51 plates)	use_cwa	41.8	38.8	13	41.9	43.0	12	7
All three	soft + replicate + offsets	42.9	43.9	13	46.5	45.3	11	9

TABLE V

STANDARD CPJUMP1 BENCHMARK ON THE 40 ELIGIBLE PLATES. PER-TRACK ROWS GIVE FRACTION RETRIEVED AT $q < 0.05$ WITH TRACK MAP IN PARENTHESES. SUMMARY FRACTIONS ARE UNWEIGHTED MEANS OVER TRACKS; PARENTHESES SUMMARY MAP IS POOLED OVER THE SCORED ROWS AND IS WEIGHTED BY TRACK SIZE. FRACTION RETRIEVED VARIES BETWEEN IDENTICAL RUNS; ONLY REPLICABILITY MAP IS DETERMINISTIC.

Track		Base	Replicate loss	Plate offsets	All three
Compound A549	short	0.353 (0.392)	0.435 (0.458)	0.327 (0.384)	0.497 (0.493)
	long	0.513 (0.526)	0.650 (0.583)	0.503 (0.513)	0.703 (0.626)
Compound U2OS	short	0.297 (0.344)	0.379 (0.404)	0.310 (0.351)	0.422 (0.439)
	long	0.369 (0.413)	0.471 (0.474)	0.395 (0.421)	0.503 (0.496)
CRISPR A549	short	0.173 (0.286)	0.271 (0.343)	0.092 (0.239)	0.340 (0.375)
	long	0.183 (0.296)	0.376 (0.356)	0.134 (0.282)	0.366 (0.385)
CRISPR U2OS	short	0.026 (0.175)	0.049 (0.207)	0.052 (0.182)	0.078 (0.232)
	long	0.170 (0.277)	0.219 (0.290)	0.160 (0.259)	0.258 (0.318)
ORF A549	short	0.006 (0.136)	0.012 (0.158)	0.012 (0.138)	0.019 (0.174)
	long	0.000 (0.128)	0.006 (0.142)	0.000 (0.126)	0.006 (0.141)
ORF U2OS	short	0.043 (0.171)	0.099 (0.223)	0.062 (0.209)	0.099 (0.221)
	long	0.006 (0.133)	0.012 (0.146)	0.012 (0.140)	0.031 (0.160)
Replicability summary (12 tracks)		0.178 (0.298)	0.248 (0.343)	0.172 (0.292)	0.277 (0.369)
Target-matching summary (8 tracks)		0.339 (0.197)	0.262 (0.160)	0.420 (0.194)	0.172 (0.148)
Gene-compound summary		0.005 (0.123)	0.000 (0.082)	0.010 (0.140)	0.029 (0.082)

TABLE VI

SHORT-TIMELINE REPLICABILITY, FRACTION RETRIEVED, SAME BENCHMARK HARNESS. CELLCLIP ROW USES THE RELEASED CHECKPOINT AND ITS OWN EXPORT PATH. SEPARATE RUNS OF A NON-DETERMINISTIC HARNESS.

Track	CellCLIP	Base	+ replicate loss	All three
Compound A549	0.366	0.353	0.435	0.497
Compound U2OS	0.353	0.297	0.379	0.422
CRISPR A549	0.020	0.173	0.271	0.340
CRISPR U2OS	0.118	0.026	0.049	0.078
ORF U2OS	0.119	0.043	0.099	0.099

F. Cached Feature Analysis

We checked whether channel identity survives the frozen backbone using 500 sites sampled with seed 42 from plate BR00116991. The reproducible diagnostic output is stored with the figure sources. A variance decomposition attributes 31.1% of variance to between-channel differences, 37.1% to between-site differences and 31.8% to the residual (Figure 2). Same-channel tokens from different sites have mean cosine

similarity 0.84, while cross-channel tokens from different sites average 0.75 (Figure 3). The two-dimensional PCA projection in Figure 4 provides a qualitative view of this channel structure.

G. Cost

For the no-offset ablation runs, `metrics.csv` records about four seconds for the training-loop phase of an epoch with 55 steps at batch size 256 on the RTX 5080. This excludes validation, data loading outside the timer and the offset-refresh pass used by offset variants, so it is not an end-to-end run-time measurement. Both backbones stay frozen; the trainable components are the CrossChannelFormer, two projection heads and one logit-scale scalar.

VII. CONCLUSION

MorphoCLIP maps Cell Painting wells and text descriptions of perturbations into a 512-dimensional space with about 14M trainable parameters on top of two frozen backbones. One model handles compounds, CRISPR knockouts and ORF overexpressions. At one seed, perturbation-level Recall@10 is

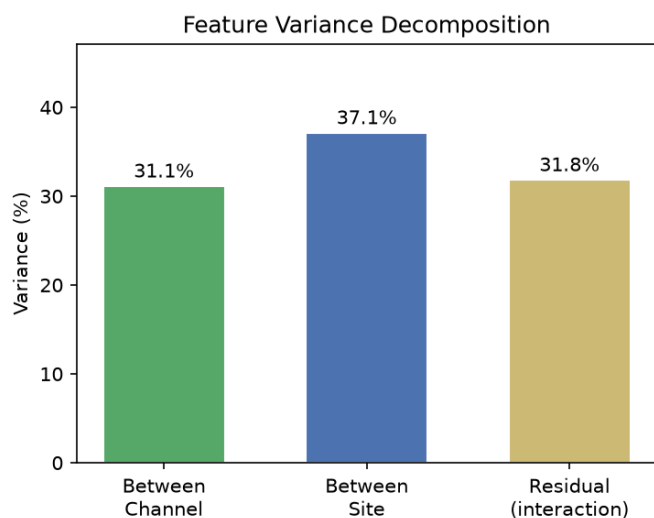


Fig. 2. Variance decomposition of the cached DINOv3 tokens on plate BR00116991: between-channel, between-site and residual components.

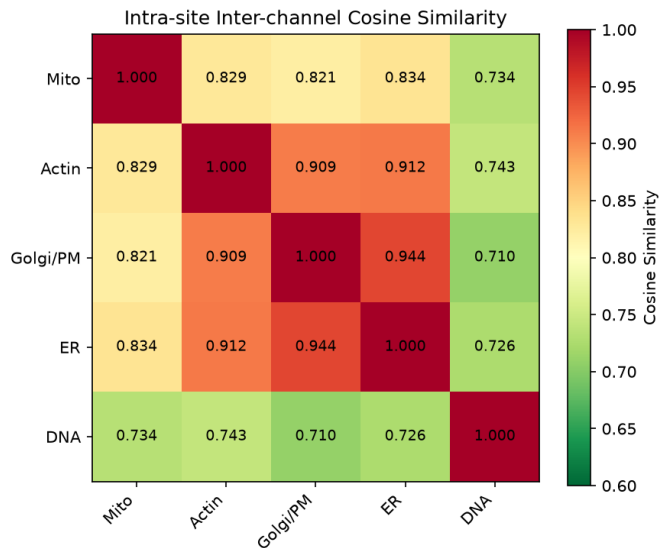


Fig. 3. Mean cosine similarity between fluorescence channels in the cached token space. The DNA channel is the most distinct.

about 39% in both directions on validation and 37 to 44% on test, against analytic chance rates of 10 to 12%. Separate stochastic benchmark runs show numerically comparable compound replicability fractions for MorphoCLIP and the released CellCLIP checkpoint but do not support a ranking. The base model’s four-track mean compound target-matching fraction lies within the published CellProfiler range, although one track is above it.

Gene-aware soft labels, the replicate alignment loss and condition-relative plate offsets each change validation retrieval by only a few perturbations, and the test split reorders them. These retrieval differences are unresolved without repeated seeds. The replicate loss does increase deterministic replicabil-

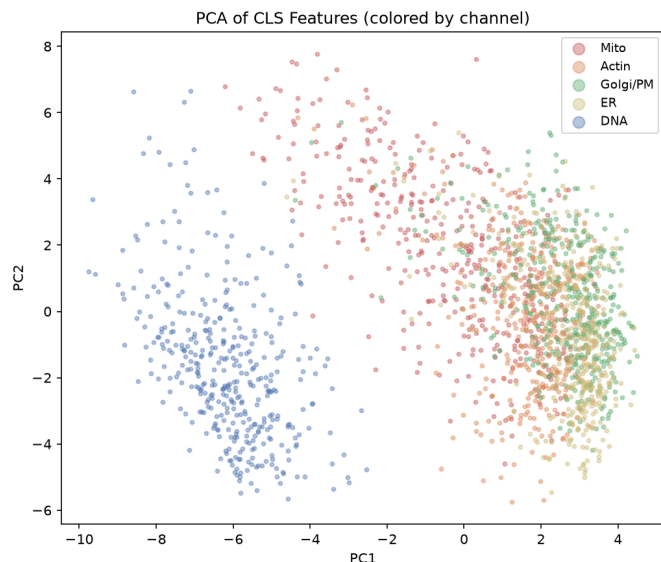


Fig. 4. First two principal components of the cached 1024-d channel tokens, colored by fluorescence channel.

ity mAP on every track, from pooled 0.298 to 0.343, while the combined run reaches 0.369. This result is partly definitional because the loss directly optimizes replicate proximity. The in-batch plate correction degrades retrieval because the plate mean carries condition information named by the prompt; the condition-relative offset avoids that degradation but shows no consistent retrieval or benchmark-mAP improvement at this seed count.

Gene-compound fraction retrieved is at or near zero for every model: at most three admitted pairs pass the stochastic significance gate in any run. We have no positive cross-modality result.

A. Limitations and Future Work

Every retrieval number in this paper is from one seed. Repeated seeds are needed for the control, replicate-loss and combined runs. Fraction retrieved on the standard benchmark varies between runs of the same code, so those values are descriptive. We did not rerun CellProfiler, and the CellCLIP row comes from a separate run with a different export path and a batch-correction step ours lacks. Because the replicate loss directly optimizes the replicability task, evidence that the embedding is more informative requires a task outside that objective; target matching does not provide such evidence here.

On cross-modality, one concrete lead is the choice of gene annotation. Training soft labels use each compound’s primary target (160 distinct genes) while the benchmark’s gene-compound track uses the full target list (758 distinct genes, off-targets included). Aligning the two, or training on the wider list, is a one-line change we have not measured. A second lead is prompt content: whether the SMILES string and gene-function text carry signal through BioClinical ModernBERT, or are inert, can be tested by ablating them out of the templates.

Beyond CPJUMP1, the full JUMP corpus of about 140,000 perturbations across many sites and cell lines is the setting in which a text-supervised model would have to prove that its embeddings transfer.

We release the code, configs, split manifests and the evaluation harness with the paper.

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